



December 2025

(For CIV)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

20 Marks

- 1-



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12. Section modulus of circular section with diameter D is equal to
 A) $\pi D^3/64$ B) $\pi D^3/32$
 C) $\pi D^3/16$ D) $\pi D^3/8$
13. A cantilever beam of length L is subjected to a concentrated load W at free end. The maximum deflection at free end is equal to
 A) $WL^3/12EI$ B) $WL^3/9EI$
 C) $WL^3/6EI$ D) $WL^3/3EI$
14. A simply supported beam of length L is subjected to a uniformly distributed load of w per unit run over the whole span. The maximum deflection is equal to
 A) $wL^4/384EI$ B) $wL^4/192EI$
 C) $5wL^4/384EI$ D) $5wL^4/192EI$
15. The maximum shear force in a simply supported beam of span ' L ' subjected to a central point load, W is given by the following expression
 A) $W/2$ B) $W/3$
 C) $W/4$ D) W
16. The shape of the bending moment diagram over the length of a beam, carrying a uniformly distributed load is always
 A) Linear B) Circular
 C) Parabolic D) Cubic Parabola
17. Radial stress in thin cylinders is
 A) Uniform B) Parabolic
 C) Neglected D) Linear
18. Which cylinder can also be subjected to external pressure
 A) Thick cylinder B) Thin cylinder
 C) Both thin and thick cylinder D) None of the above
19. What is the effective length of a column at both ends hinged?
 A) $2L$ B) $L/3$
 C) $L/2$ D) L
20. What is the effective length of a column with one end fixed and another end free?
 A) $2L$ B) $L/3$
 C) $L/2$ D) L

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

1. a) Define stress and strain. Draw a neat sketch of stress strain diagram for mild steel and show salient points.
 b) The state of stress at a point in a strained material is shown in Figure 1 (b). Determine i) Principal stresses and their planes ii) maximum shear stresses and their planes.

Marks	BT*	CO*	PO*
08	L*1	1	1

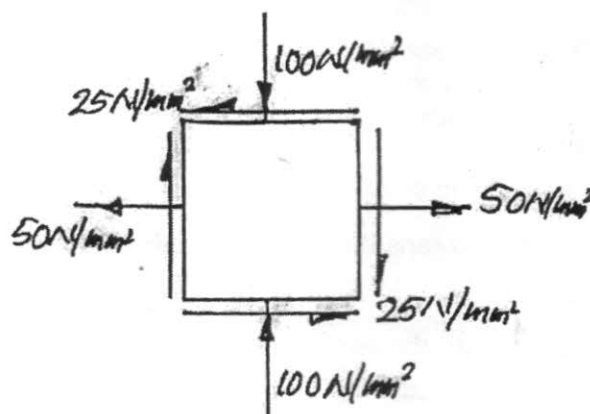


Figure 1 (b)

08	L3	2	1,2
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2. a) A block 50 mm x 50 mm x 50 mm is subjected to compressive load of 100 kN on all the three faces. If Poisson's ratio is 0.3 and E is 200 Gpa determine the change in length, breadth, depth and volume.
- b) The state of stress at a point in a strained material is shown in Figure 2 (b). Determine i) Principal stresses and their planes ii) Normal and shear stress on plane AE.

08 L3 1 1,2

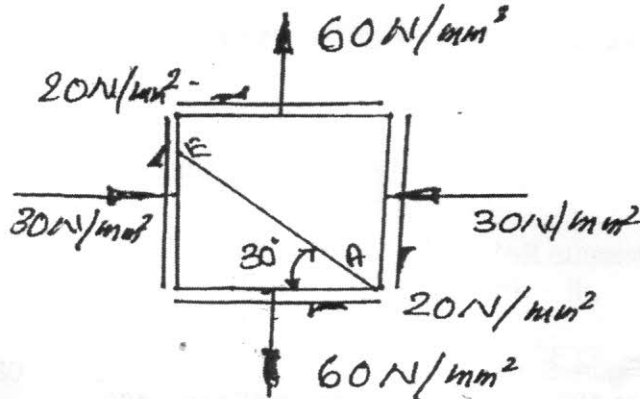


Figure 2 (b)

08 L3 2 1,2

3. a) A steel rod 8.0 m long is at a temperature of 20°C. Find the stresses in the bar if the temperature of bar is increased to 65°C for i) Expansion of the rod is prevented completely ii) The rod is permitted to expand by 2.0 mm. Take $\alpha = 12 \times 10^{-6}/^\circ\text{C}$ and $E = 205$ Gpa.
- b) The state of stress at a point in a strained material is shown in Figure 3 (b). Determine i) Principal stresses and their planes ii) maximum shear stresses and their planes.

08 L3 1 1

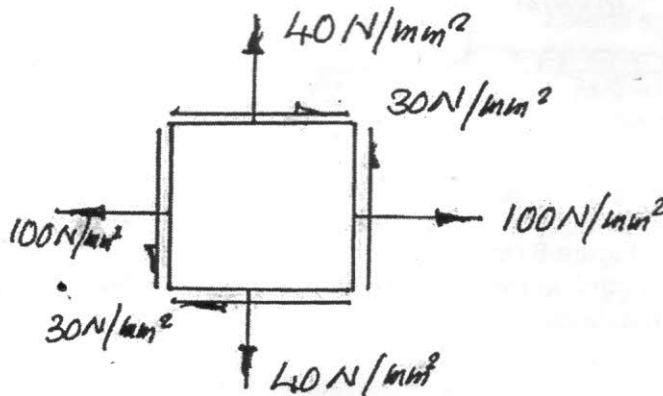


Figure 3 (b)

08 L3 2 1,2

Unit – II

4. a) A beam AB is loaded as shown in Figure 4 (a). Draw SFD and BMD showing salient features.

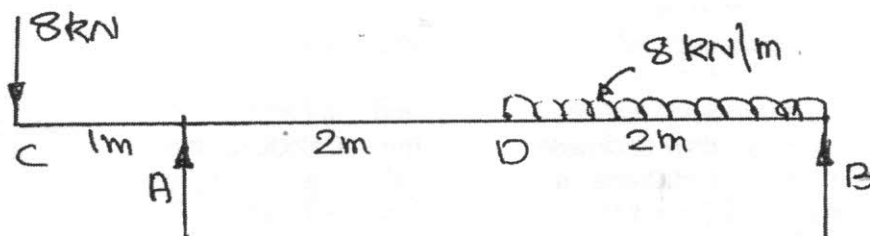


Figure 4 (a)

08 L3 3 1,2

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- b) Prove that in the case of solid circular section, the maximum shear stress is $4/3$ times the average shear stress.

08 2 4 1

5. a) A beam AB is loaded as shown in Figure 5 (a). Draw SFD and BMD showing salient features.

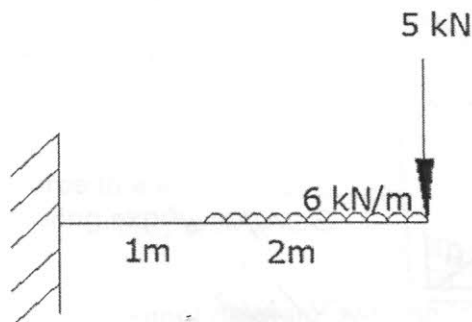


Figure 5 (a)

08 L3 3 1,2

- b) A symmetrical I section with flange 100 mm x 15 mm and web 15 mm x 100 mm is subjected to a bending moment of 50 kN – m. Sketch the variation of bending stress across the depth.

08 3 4 3

6. a) A beam AB is loaded as shown in Figure 6 (a). Draw SFD and BMD showing salient features.

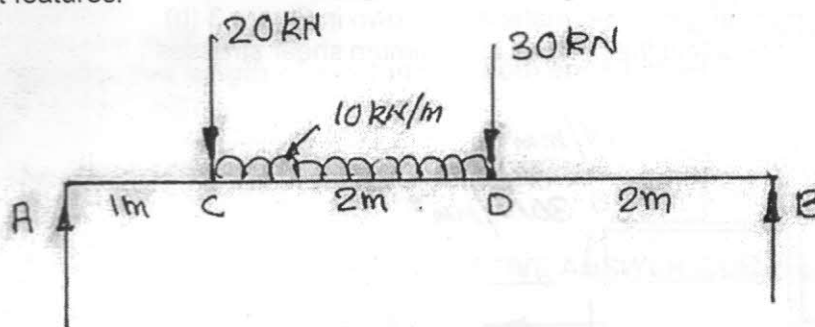


Figure 6 (a)

08 L3 3 1,2

- b) A simply supported beam of length L is subjected to udl W/m for the entire span. Determine the maximum deflection.

08 L3 4 2

Unit – III

7. a) A cast iron column with square cross section of side 50mm is 3.0 m long and fixed at both ends. Calculate the safe load by Rankine's formula using factor of safety of 3. Take $\sigma_c = 550 \text{ N/mm}^2$ and $\alpha = 1/1600$.

08 L3 5 1,2

- b) Determine the maximum and minimum hoop stresses across the section of pipe of 200 mm internal diameter and 100 mm thick, when the pipe contains a fluid at a pressure of 20 N/mm². Also draw the variation of radial pressure and hoop stress across the thickness.

08 L3 5 2

8. a) Define effective length in columns. Give the values of effective lengths based on end conditions.

08 L2 5 1,2

- b) A cylindrical thin shell of 4.0 m long with closed at both ends has an internal diameter of 1.0 m and a wall thickness of 12 mm. Calculate the circumferential and longitudinal stresses induced and also the change in volume of the shell if it is subjected to an internal pressure of 2 N/mm². Assume E as 200 Gpa and Poisson's ratio as 0.25.

08 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
